Object-Oriented Programming (CS112)

Lab Task-10

[CLO-2, Taxonomy Level-C3, GA3]

Course: BSSE-II **Semester:2nd (**Spring 2025)

Instructions

Plagiarism, copy & paste material will lead to the cancellation of your assignment. The input values should be different, don't copy the solution from external source as the plagiarism will be tested through Turnitin and the plagiarized assignments will lead to marks' deduction.

Objectives:

- Event handling in java
- Operations on clicks
- Event handler/Action listener
- Interface

Set of implemented code already available is an interface, like an action listener. To add event you have to implement this into your program

EVENT HANDLING IN JAVA

- **Eventhandler** and **ActionListener** are the two important aspects of event manipulation
- Adding Listener helps to organize the action(event) to be done after the button or the GUI control is selected
- **ActionListner** is the main Interface in event handling

Need to add: import java.awt.event.ActionEvent; import java.awt.event.ActionListener;

ComboBox: Provides you with a list of items

```
String[] roles = {"Manager", "Eng Lead", "Eng Designer", "Jr Designer"};
JComboBox roleComboBox = new JComboBox(roles);
```

For table creation, need to create array for row and columns then pass this to table model

If table is large you can add scrollpane to support scrolling, additionally use colorpane for adding background and foreground colors.

```
// Create the table for displaying data
String[] columnNames = {"FirstName", "LastName", "Id Number"};
Object[][] data = {
   {"John", "Doe", "101"},
   {"Jane", "Smith", "102"},
   {"Mike", "Johnson", "103"},
   {"Emily", "Davis", "104"},
   {"Chris", "Brown", "105"}
};
DefaultTableModel tableModel = new DefaultTableModel(data, columnNames);
JTable table = new JTable(tableModel);
JScrollPane tableScrollPane = new JScrollPane(table);
```

Activity.1. ***Write a program to add event to a button, on click it should display "button clicked"***

Solution:

```
package gui2;
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.awt.event.ActionListener;

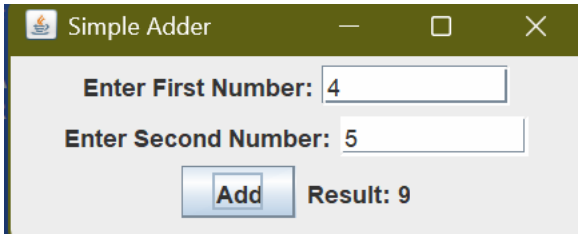
public class  PanelClass extends Frame implements ActionListener{
        public static void main(String[] args) {

   JFrame jf = new JFrame();
   jf.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
   jf.setSize(200, 300);
   JButton button = new JButton("clicknow");
   button.addActionListener(new ActionListener() {
      @Override
      public void actionPerformed(ActionEvent e) {
         System.out.println("Button Clicked!");
      }
   });
   jf.add(button);
   jf.setVisible(true);
```

```
        }
}
```

Activity.2. Repeat the above operation and show a dialog box on button click.

Q.2. Create following GUI



Solution:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

public class GUIEventDriven {
    public static void main(String[] args) {
        // Create the frame
        JFrame frame = new JFrame("Simple Adder");

        // Set layout for the frame
        frame.setLayout(new FlowLayout());

        // Create components
        JLabel label1 = new JLabel("Enter First Number:");
        JTextField textField1 = new JTextField(10);

        JLabel label2 = new JLabel("Enter Second Number:");
        JTextField textField2 = new JTextField(10);
```

```
        JButton addButton = new JButton("Add");
        JLabel resultLabel = new JLabel("Result: ");

        // Add components to the frame
        frame.add(label1);
        frame.add(textField1);
        frame.add(label2);
        frame.add(textField2);
        frame.add(addButton);
        frame.add(resultLabel);

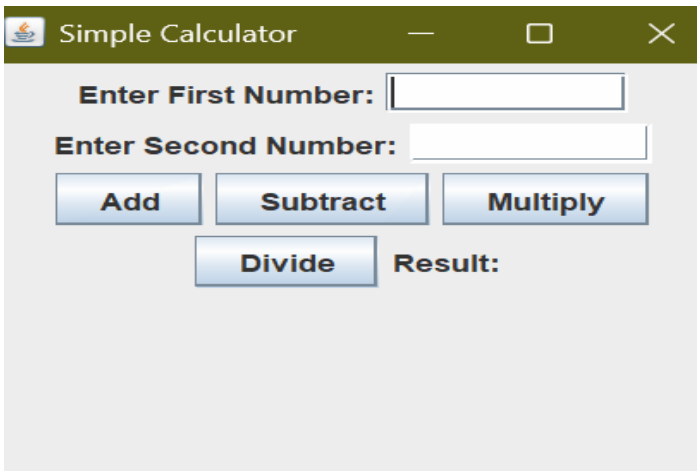
        // Set action listener for the button
        addButton.addActionListener(new ActionListener() {
            @Override
            public void actionPerformed(ActionEvent e) {
                // Get inputs and add them
                int num1 = Integer.parseInt(textField1.getText());
                int num2 = Integer.parseInt(textField2.getText());
                int sum = num1 + num2;

                // Display result
                resultLabel.setText("Result: " + sum);
            }
        });

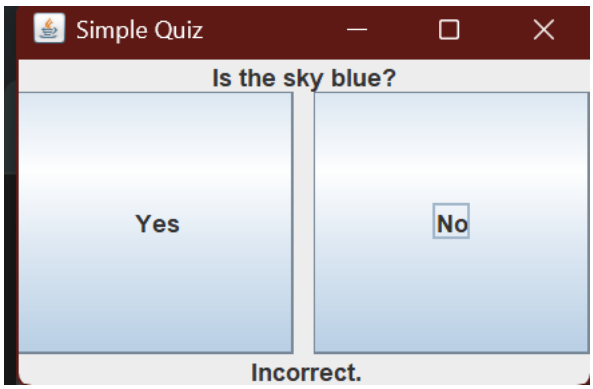
        // Set frame properties
        frame.setSize(300, 150);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```

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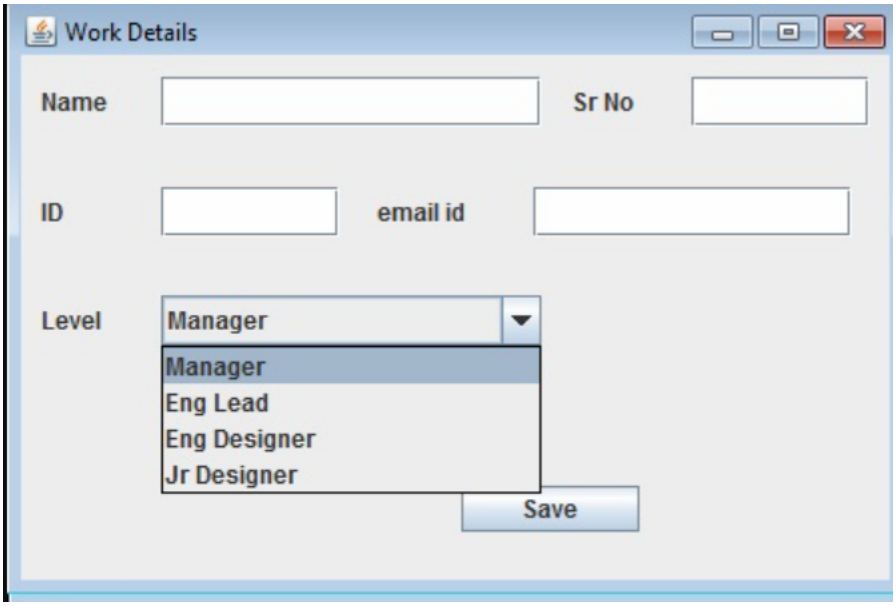
1. Create the following GUI with title “Simple Calculator” having the following components. Add events behind each button and show result in label.



2. Create a Java Swing application that displays a single **question** with **two answer buttons** ("Yes" and "No") arranged side by side using GridLayout. When the user clicks either button, show a message below the buttons indicating whether the answer is correct or not. Use JLabel to show the question and feedback, JButton for the answers, and ActionListener to handle the button clicks.



3. Create the following GUI in Java, upon save show a message ‘data is saved’ on console.



4. Create the following GUI and upon button click show a simple message on console.

